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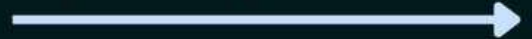
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Most Asked Interview Questions

# 50 SELENIUM INTERVIEW QUESTIONS...

*Super Important*



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## Q1. What is Selenium?

Selenium is an open-source automation testing tool for web applications across different browsers and platforms.

## Q2. What are the different components of Selenium?

Selenium has four main components:

- Selenium IDE
- Selenium RC (deprecated)
- Selenium WebDriver
- Selenium Grid

## Q3. What are the advantages of Selenium?

- Open-source and free
- Supports multiple browsers and platforms
- Supports various programming languages (Java, Python, C#, etc.)
- Strong community support
- Integration with other tools like TestNG, JUnit, Maven, etc.



## Q4. What are the limitations of Selenium?

Cannot test desktop applications

- No built-in reporting
- Limited support for handling CAPTCHA and images
- Requires external tools for handling mobile testing

## Q5. What is Selenium WebDriver?

Selenium WebDriver is a web automation framework that allows you to execute cross-browser tests.

## Q6. What is the difference between Selenium IDE and WebDriver?

- Selenium IDE is a record-and-playback tool that's simpler but limited.
- WebDriver allows advanced scripting and supports multiple browsers, programming languages, & frameworks.

## Q7. What browsers does Selenium support?

Selenium supports major browsers like Chrome, Firefox, Safari, Edge, and Internet Explorer.



## Q8. What is the difference between `findElement()` and `findElements()`?

- **`findElement()`**: Returns a single `WebElement`.
- **`findElements()`**: Returns a list of `WebElements` or an empty list if no elements are found.

## Q9. Can Selenium be used for mobile testing?

Selenium itself cannot test mobile apps directly, but tools like Appium extend Selenium WebDriver functionality to mobile platforms.

## Q10. What is Selenium Grid?

Selenium Grid allows parallel test execution on multiple machines and browsers to speed up the testing process.

## Q11. What is the difference between implicit and explicit waits?

- **Implicit Wait**: Applies to all elements and sets a default wait time if an element is not found immediately.
- **Explicit Wait**: Applies to specific conditions or elements.



### Q12. What is Fluent Wait in Selenium?

Fluent Wait is a type of explicit wait where you can set the polling frequency and conditions for waiting, allowing customization beyond default waits.

### Q13. What is a headless browser?

A headless browser performs the same functionality as a standard browser but without the graphical user interface (GUI). Common examples include Chrome headless and PhantomJS.

### Q14. Can Selenium handle windows-based popups?

Selenium cannot directly handle windows-based popups because it's a web-based automation tool. Third-party tools like Autolt or Robot class are needed.

### Q15. How can you handle JavaScript pop-ups in Selenium?

JavaScript alerts, prompts, and confirmation boxes can be handled using Selenium's `switchTo().alert()` method.

### Q16. What is the difference between a window handle and a window title?

- **Window Handle:** A unique identifier assigned to each window opened by the browser.
- **Window Title:** The name displayed in the title bar of the browser window.



### Q17. What is the Page Object Model (POM)?

POM is a design pattern that creates an object repository for web elements, allowing tests to be more maintainable and readable.

### Q18. What is Page Factory in Selenium?

Page Factory is an implementation of the POM, providing annotations like `@FindBy` to initialize web elements easily.

### Q19. How do you take a screenshot in Selenium?

Selenium WebDriver provides the `TakesScreenshot` interface for capturing screenshots.

### Q20. What are some common exceptions in Selenium?

- `NoSuchElementException`
- `TimeoutException`
- `ElementNotVisibleException`
- `WebDriverException`
- `StaleElementReferenceException`

### Q21. What is the purpose of `driver.quit()` and `driver.close()`?

- **`driver.quit()`**: Closes all browser windows opened by the WebDriver.
- **`driver.close()`**: Closes the current browser window.

### Q22. How can you switch between different browser windows or tabs?

- Use the **`getWindowHandles()`** method to get all the window handles and then switch using **`switchTo().window(handle)`**.



### Q23. What is the difference between `isDisplayed()`, `isEnabled()`, and `isSelected()`?

- **`isDisplayed()`**: Checks if the element is visible on the page.
- **`isEnabled()`**: Checks if the element is enabled for interaction.
- **`isSelected()`**: Checks if a checkbox or radio button is selected.

### Q24. How do you handle frames in Selenium?

Use the `switchTo().frame()` method to handle frames and move between them.

### Q25. What is an `iframe`, and how do you handle it?

An `iframe` is an inline frame used to embed another document within the current HTML document. Handle `iframes` using `switchTo().frame()`.

### Q26. What are some Selenium alternatives for browser automation?

Alternatives include Cypress, Puppeteer, TestCafe, and Playwright.

### Q27. Can Selenium be integrated with test management tools?

Yes, Selenium can be integrated with tools like TestNG, JUnit, Jenkins, and others for reporting, test case management, and CI/CD.

### Q28. What is the use of `Actions` class in Selenium?

- The `Actions` class allows you to simulate keyboard and mouse events such as drag-and-drop, hover, and right-click



### Q29. What is XPath?

XPath is a query language used to select nodes from an XML or HTML document.

### Q30. What is the difference between absolute and relative XPath?

- **Absolute XPath:** Starts from the root node and uses a full path (/html/body/div).
- **Relative XPath:** Starts from the middle of the DOM structure (//div[@id='example']).

### Q31. What are the different types of waits in Selenium?

- Implicit Wait
- Explicit Wait
- Fluent Wait

### Q32. How can you maximize a browser window in Selenium?

Use `driver.manage().window().maximize()` to maximize the browser window.

### Q33. How do you run tests in multiple browsers using Selenium?

Use WebDriver's interface for different browsers (ChromeDriver, FirefoxDriver, etc.), or use tools like TestNG with browser parameters for cross-browser testing.

### Q34. What are WebElements in Selenium?

- WebElements represent elements on a web page (e.g., buttons, input fields) that Selenium can interact with.



### Q35. What is the use of `get()` and `navigate().to()`?

- **`get()`**: Navigates to a specified URL and waits for the page to load completely.
- **`navigate().to()`**: Also navigates to a URL but doesn't wait for the full page load.

### Q36. How do you perform a double-click action in Selenium?

Use the Actions class and call `doubleClick()` on the WebElement.

### Q37. What is the role of TestNG in Selenium?

TestNG is a testing framework integrated with Selenium for managing test cases, running parallel tests, generating reports, and handling assertions.

### Q38. How can you upload files in Selenium?

You can simulate file upload by using the `sendKeys()` method to input the file path into an upload field.

### Q39. What are some common strategies to locate elements in Selenium?

- ID
- Name
- Class Name
- Tag Name
- Link Text / Partial Link Text
- CSS Selector
- XPath



#### Q40. What is the difference between `getText()` and `getAttribute()`?

- **`getText()`**: Retrieves the visible inner text of a `WebElement`.
- **`getAttribute()`**: Retrieves the value of a specific attribute of the `WebElement`.

#### Q41. Can Selenium handle HTTPS certification issues?

Yes, Selenium can handle these issues using browser-specific settings or `DesiredCapabilities`.

#### Q42. What is the role of `JavascriptExecutor` in Selenium?

`JavascriptExecutor` allows executing JavaScript code within the browser, which can be useful for handling dynamic elements or scrolling.

#### Q43. What is the importance of `DesiredCapabilities` in Selenium?

`DesiredCapabilities` is used to set browser properties like browser name, version, platform, etc., before starting the `WebDriver` session.

#### Q44. What is the difference between `WebDriver` and `Remote WebDriver`?

- **`WebDriver`**: Local browser automation.
- **`Remote WebDriver`**: Automates a browser on a remote server (used in Selenium Grid).

#### Q45. What is a proxy in Selenium, and how do you use it?

A proxy server can be used to route traffic through a middle layer for testing purposes. You can configure proxies using `DesiredCapabilities`.



#### Q46. What is a WebDriver Wait in Selenium?

WebDriverWait is an explicit wait applied to wait for a specific condition or element before throwing an exception.

#### Q47. Can you automate Captcha using Selenium?

No, Captcha is designed to prevent automation, so you cannot automate it using Selenium.

#### Q48. What is the difference between `navigate().refresh()` and `driver.get(driver.getCurrentUrl())`?

- **`navigate().refresh()`**: Refreshes the browser window.
- **`driver.get(driver.getCurrentUrl())`**: Loads the current URL again, which also results in a page refresh.

#### Q49. What are Selenium's advantages over QTP/UFT?

- Open-source (free)
- Supports multiple programming languages
- Cross-platform
- Larger community support

#### Q50. Can Selenium automate desktop applications?

- No, Selenium can only automate web applications. Desktop automation requires tools like Autolt or Robot Framework.
- This list will help you navigate key concepts around Selenium without touching on any coding-related aspects.